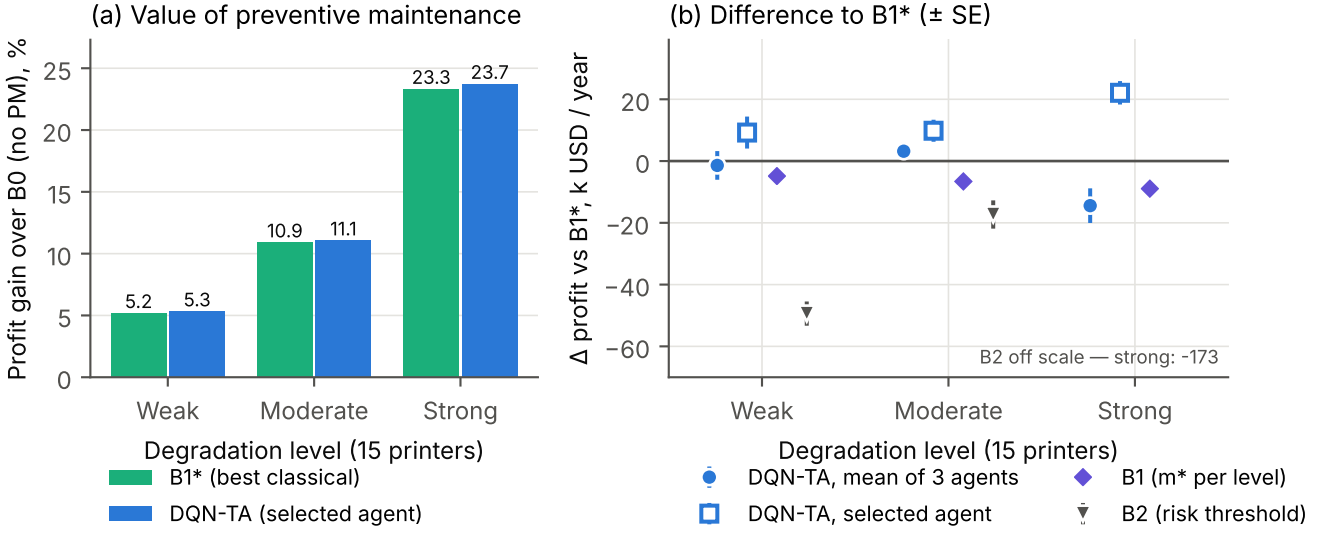
(a) Value of preventive maintenance
Profit gain over B0 (no PM), %
25
20
15
10
5
0
5.2
5.3
10.9
11.1
23.3
23.7
Weak
Moderate
Strong
Degradation level (15 printers)
B1* (best classical)
DQN-TA (selected agent)
(b) Difference to B1* (± SE)
Δ profit vs B1*, k USD / year
20
0
−20
−40
−60
B2 off scale — strong: -173
Weak
Moderate
Strong
Degradation level (15 printers)
DQN-TA, mean of 3 agents
DQN-TA, selected agent
B1 (m* per level)
B2 (risk threshold)